

Citehouse turns a UK company name or Companies House number into an explainable, citation-backed intelligence report based on the following pillars

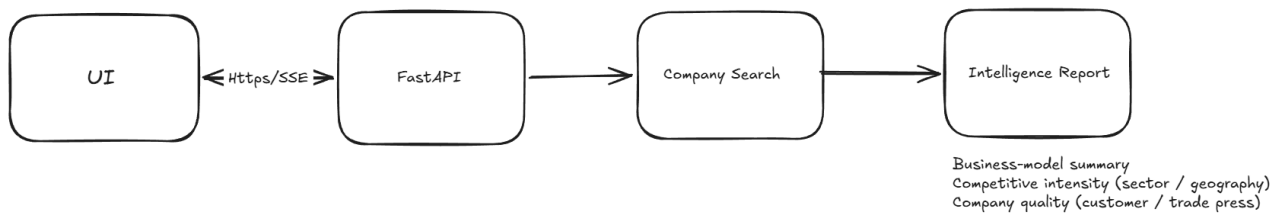
Report pillars

1. Business-model summary
2. Competitive intensity (sector / geography)
3. Company quality (customer / trade press)

Stack

- FastAPI - Backend (Wide support, Pydantic Validation Support)
- Nextjs + Shadcn - FrontEnd

High-level system



User Flow

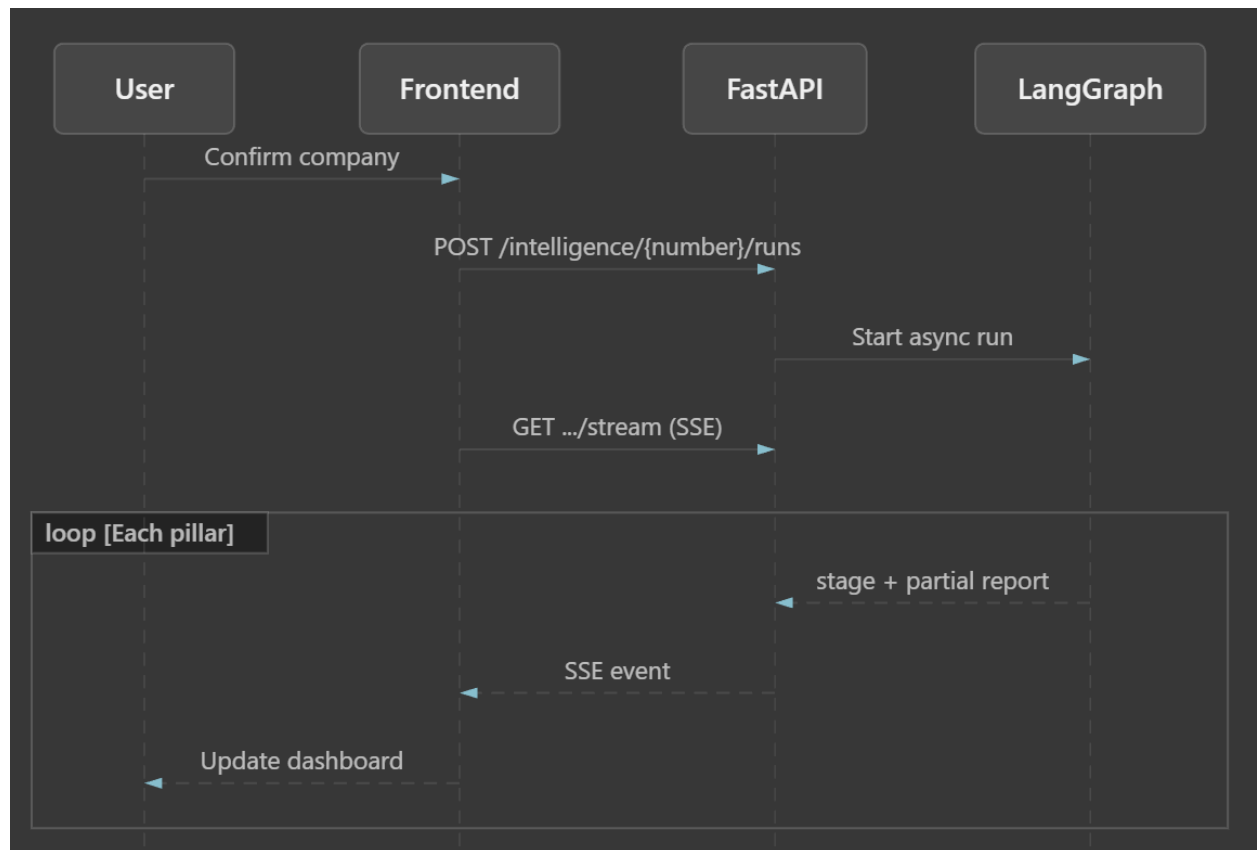
1. Find the company



Tools: `search_companies`, `filter_companies`, `get_company`.

Max 5 search attempts so the agent can retry before giving up.

2. Build the report



Dashboard order: **Profile** → **Business model** → **Competition** → **Quality**.

Agent orchestration

Citehouse deploys 2 agentic workflows

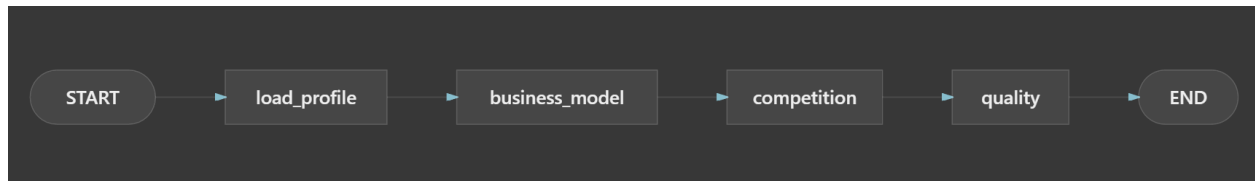
1. For Searching companies (Company-search agent)
2. For generating the report (Report Generation Agent)

Company-search agent



Framework	LangGraph <code>create_react_agent</code>
Output	<code>found / needs_clarification / not_found</code>
Sources	Query Companies House (CH) Public Data API https://api.company-information.service.gov.uk

Report Generation Agent



Node	Tools	Output
load_profile	CH profile (deterministic)	CompanyIdentity
business_model	get_company_profile	BusinessModelSection
competition	profile, peers, web search	CompetitionSection
quality	web search (+ profile context)	QualitySection

Design choices

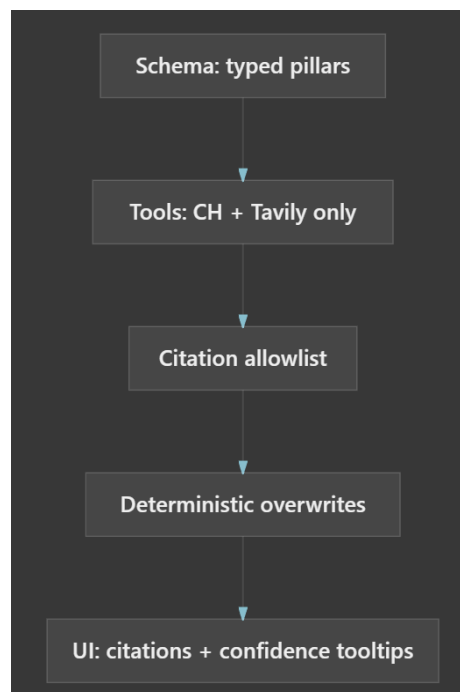
- Sequential so later pillars can use earlier structured context.
- After each agent, deterministic code overwrites peers, confidence, and citations.
- SSE streams pillar results as they finish.

Data sources

Source	Used for
Companies House	Profile, search, peers
Tavily (Web search)	Competition / quality snippets + URLs
OpenAI (Via Langchain)	Agents + structured sections

Grounding

Evidence is constrained in layers — schema → tools → post-filters → UI.



Citation allowlist (`citations.py`)

1. Web URLs must match that run's Tavily results.
2. CH refs must be known paths (e.g. `companies_house:profile.sic_codes`).
3. Invalid citations are stripped; a gap note records removals.

This proves the source is correct, not that every sentence is entailed by a citation.

Deterministic overwrites

Field	Source of truth
Peer set / count / arena	Fresh <code>search_peers</code>
Pillar confidence	Rules in <code>confidence.py</code>
Confidence factors	Boolean ticks for UI tooltips

Uncertainty Handling

Pillar	Higher confidence when...	Low when...
Business model	SIC codes present → medium	No SIC
Competition	Peers + web name match + address corroboration → high; peers only → medium	No peers
Quality	Attributable Trustpilot review page and trade/news URL naming the company, plus profile corroboration → medium	Either site missing

UI shows ✓ or unmet criteria in confidence tooltips. free-text summaries are still Model Judged

Trade-offs

Decision	Benefit	Cost
CH-only for business model	No dedicated websites fetch	Less product/market detail
Sequential pillars	Clear abstraction	Higher latency than parallel
Citation allowlist	Blocks invented sources	No actually logical check of the quoted source
In-memory + file runs	Fast local demo	No persistence / shared store